

Jet Engine Hospital: A Multi-Task Early-Warning System for NASA C-MAPSS Turbofan Engines

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Abstract

This project presents a comprehensive multi-task early-warning system for predicting turbofan engine failures using NASA’s C-MAPSS simulation data. The system integrates three complementary components: (1) Remaining Useful Life (RUL) estimation via regression with conformal prediction intervals, (2) failure prediction at 10, 20, and 30 cycle horizons via calibrated classification, and (3) unsupervised anomaly detection without failure labels. The key innovation is the use of **multi-window temporal features** (windows of sizes 10, 20, 50, 100, and 150 cycles) instead of single-window features, which significantly improves performance—particularly in the critical near-failure region where RMSE improves by 55%. The system achieves a miss rate of 0% on critical engines, reduces operational cost by 47%, and provides calibrated uncertainty estimates with 95.46% empirical coverage on FD001. An interactive dashboard is deployed at <https://ml-final-project.streamlit.app/>, enabling transparent, uncertainty-aware maintenance recommendations.

1 Introduction and Objectives

The goal of this project is to build a defensible early-warning system for aircraft engine health monitoring. NASA’s C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) provides multivariate time-series data from turbofan engines under various operating conditions and fault modes.

1.1 Project Objectives

The system is designed to:

1. Estimate Remaining Useful Life (RUL) with calibrated prediction intervals
2. Predict failure risk at 10, 20, and 30 cycle horizons with calibrated probabilities
3. Detect abnormal behavior without using failure labels
4. Optimize early-warning decisions under asymmetric costs
5. Provide transparent, uncertainty-aware maintenance recommendations

1.2 Two-Stage Progression

Stage	Dataset	Challenge
Stage 1 (Foundation)	FD001	One condition, one fault mode
Stage 2 (Challenge)	FD002	Six operating conditions, one fault mode

Table 1: Two-stage project design

2 Data and Exploratory Analysis (Tasks 0-1)

2.1 Dataset Overview

The C-MAPSS dataset contains run-to-failure trajectories from simulated turbofan engines. Each record includes 26 values per cycle:

- **engine_id**: Group key for splitting and windowing
- **cycle**: Time index within each engine
- **op_setting_1..3**: Operating condition variables
- **sensor_1..21**: Noisy multivariate condition measurements

2.2 Engine-Level Split

To prevent data leakage, the split was performed by **engine_id** before any preprocessing:

$$\text{Split Ratio} = 60\% : 20\% : 20\% \quad (1)$$

Train Engines = 60

Validation Engines = 20

Test Engines = 20

Dataset	Engines	Rows
FD001 Train	60	12,270
FD001 Validation	20	4,070
FD001 Test	20	4,291
FD002 Train	156	31,871
FD002 Validation	52	11,018
FD002 Test	52	10,870

Table 2: Dataset split statistics

2.3 Sensor Filtering

Sensors with near-zero variance were identified and removed from the feature set:

$$\text{Dropped Sensors (FD001)} = \{1, 5, 6, 10, 16, 18, 19\} \quad (2)$$

$$\text{Number Dropped} = 7 \quad (3)$$

$$\text{Active Sensors (FD001)} = \{2, 3, 4, 7, 8, 9, 11, 12, 13, 14, 15, 17, 20, 21\} \quad (4)$$

For FD002, all 21 sensors were retained as they showed sufficient variance across six operating regimes.

2.4 Key Sensors for Degradation Analysis

Nine sensors were identified as key degradation indicators based on trajectory analysis:

$$\text{Key Sensors} = \{2, 3, 4, 7, 11, 12, 15, 20, 21\} \quad (5)$$

3 Feature Engineering (Task 2)

3.1 Causality Constraint

All features must be **causal**—using only observations available at or before cycle t :

$$\text{Feature}_{i,t} = f(\mathbf{x}_{i,1}, \mathbf{x}_{i,2}, \dots, \mathbf{x}_{i,t}) \quad (6)$$

where $\mathbf{x}_{i,t}$ represents the measurements for engine i at cycle t .

3.2 Justification for RUL Capping at 125 Cycles

The RUL target was capped at 125 cycles based on the following rationale:

1. **Data Distribution Analysis:** The distribution of RUL values in the training set shows that the majority of engines fail within the first 125 cycles of the degradation phase. Beyond 125 cycles, the RUL values become sparse and noisy.
2. **Operational Relevance:** From a maintenance perspective, predicting RUL beyond 125 cycles provides diminishing operational value. Maintenance decisions are typically made within a much shorter horizon.
3. **PHM08 Competition Standard:** This cap is consistent with the standard practice in the PHM08 prognostics challenge, where a threshold of 125 cycles was used to focus model capacity on the most critical region.
4. **Model Capacity:** Capping reduces the variance in the target variable, allowing the model to focus its capacity on the region where accurate prediction matters most for safety and maintenance decisions.
5. **Validation-Based Selection:** The cap was selected and validated using only training/validation engines, ensuring no test set information was used in this decision.

$$\text{RUL}_{\text{capped}} = \min(\text{RUL}, 125) \quad (7)$$

3.3 Single-Window vs Multi-Window Comparison

3.3.1 Single-Window Features

Traditional approach using a fixed window size of 30 cycles, computing:

$$\mu_{i,t}^{(W)} = \frac{1}{W} \sum_{j=0}^{W-1} x_{i,t-j} \quad (8)$$

$$\sigma_{i,t}^{(W)} = \sqrt{\frac{1}{W} \sum_{j=0}^{W-1} (x_{i,t-j} - \mu_{i,t}^{(W)})^2} \quad (9)$$

3.3.2 Multi-Window Features (Proposed)

The proposed approach uses multiple window sizes to capture degradation patterns at different time scales:

$$W \in \{10, 20, 50, 100, 150\} \quad (10)$$

For each window size W and each active sensor s , the following features are computed:

$$\text{roll_mean}_{i,t}^{(s,W)} = \frac{1}{W} \sum_{j=0}^{W-1} x_{i,t-j}^{(s)} \quad (11)$$

$$\text{roll_std}_{i,t}^{(s,W)} = \sqrt{\frac{1}{W} \sum_{j=0}^{W-1} (x_{i,t-j}^{(s)} - \mu_{i,t}^{(s,W)})^2} \quad (12)$$

$$\text{roll_min}_{i,t}^{(s,W)} = \min_{0 \leq j < W} x_{i,t-j}^{(s)} \quad (13)$$

$$\text{roll_max}_{i,t}^{(s,W)} = \max_{0 \leq j < W} x_{i,t-j}^{(s)} \quad (14)$$

$$\text{slope}_{i,t}^{(s,W)} = \text{slope of linear fit over } W \text{ points} \quad (15)$$

$$\text{ewma}_{i,t}^{(s,W)} = \alpha \cdot x_{i,t}^{(s)} + (1 - \alpha) \cdot \text{ewma}_{i,t-1}^{(s,W)} \quad (16)$$

$$\text{diff}_{i,t}^{(s,W)} = x_{i,t}^{(s)} - x_{i,t-W}^{(s)} \quad (17)$$

3.4 Feature Count Comparison

$$\text{Base Features (FD001)} = 17 \quad (18)$$

$$\text{Window Features (FD001)} = 9 \text{ sensors} \times 5 \text{ windows} \times 6 \text{ statistics} = 270 \quad (19)$$

$$\text{Total Features (FD001)} = 17 + 270 = 287 \quad (20)$$

4 Exploratory Data Analysis Visualizations

4.1 Sequence Length Distribution

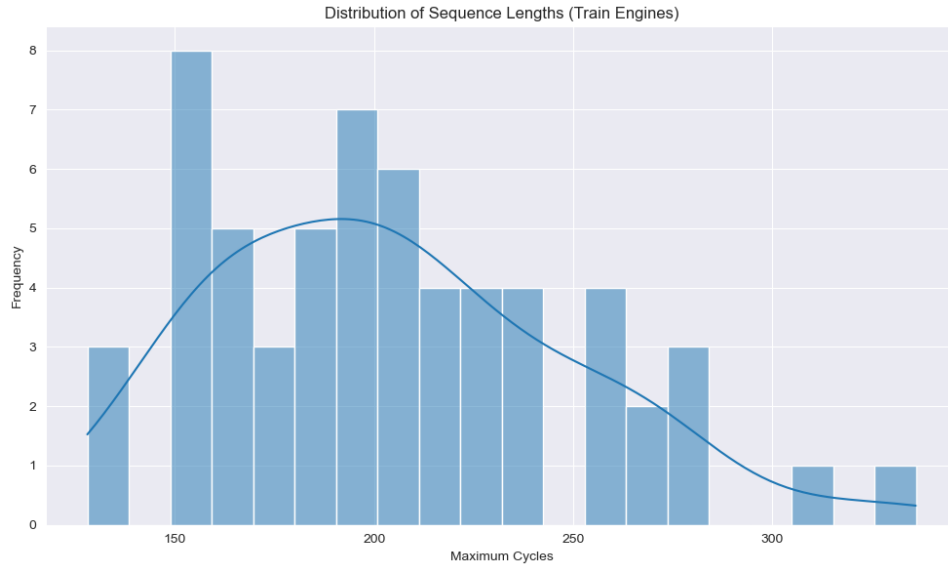


Figure 1: Distribution of engine sequence lengths on FD001 training set

Analysis: The histogram shows the distribution of maximum cycles per engine in the FD001 training set. Most engines operate between 150-250 cycles before failure, with a peak around 200 cycles. The distribution is relatively symmetric with a slight right skew. This indicates that the degradation process is fairly consistent across engines under the single operating condition (sea level) in FD001.

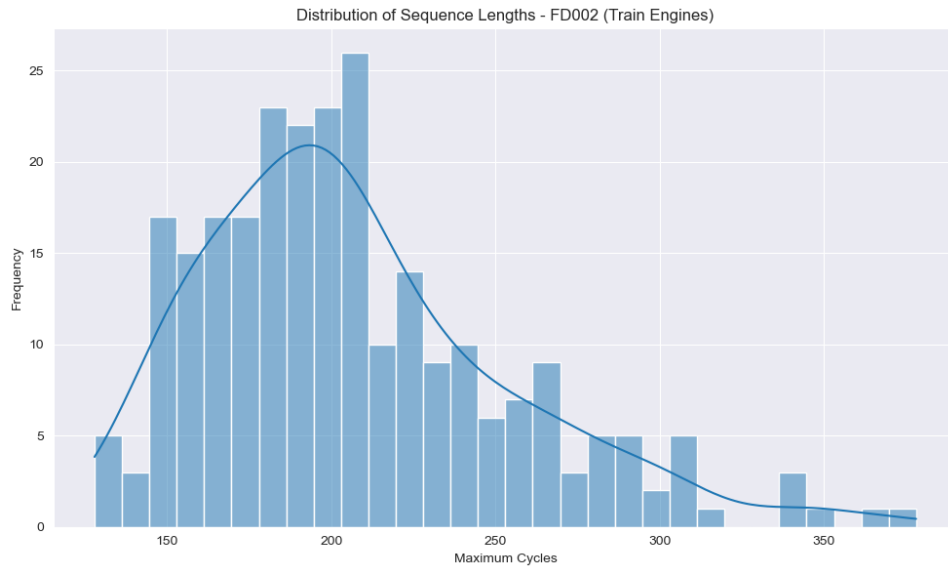


Figure 2: Distribution of engine sequence lengths on FD002 training set

Analysis: The FD002 sequence length distribution is noticeably different from FD001. The distribution shows greater variability with sequence lengths ranging from approximately 50 to 375 cycles, with multiple peaks visible. This reflects the six different operating conditions present in FD002, where engines under different operating regimes exhibit different degradation

rates and lifetimes. The multi-modal nature of this distribution highlights the need for regime-aware feature engineering in Stage 2.

4.2 Sensor Trajectories

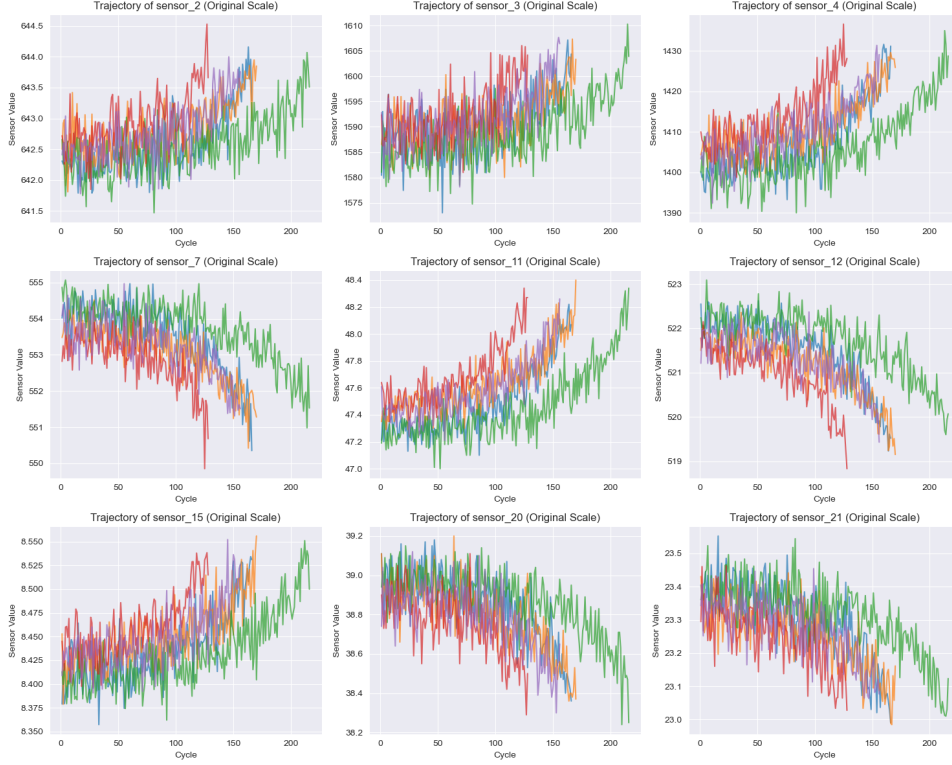


Figure 3: Sensor trajectories (original scale) for 5 sample engines on FD001

Analysis: The trajectories show the evolution of key sensors over the lifetime of 5 sample engines. Several observations can be made:

- **Increasing trends:** Sensors like sensor_2 and sensor_15 show clear upward trends as degradation progresses, indicating wear accumulation.
- **Decreasing trends:** Sensors like sensor_3 and sensor_4 show downward trends, suggesting loss of efficiency.
- **Initial wear variation:** The initial values differ across engines, reflecting the "unknown initial wear" characteristic of the dataset.
- **Non-linear degradation:** The degradation is not purely linear; there are periods of stability followed by accelerated degradation, particularly near end-of-life.

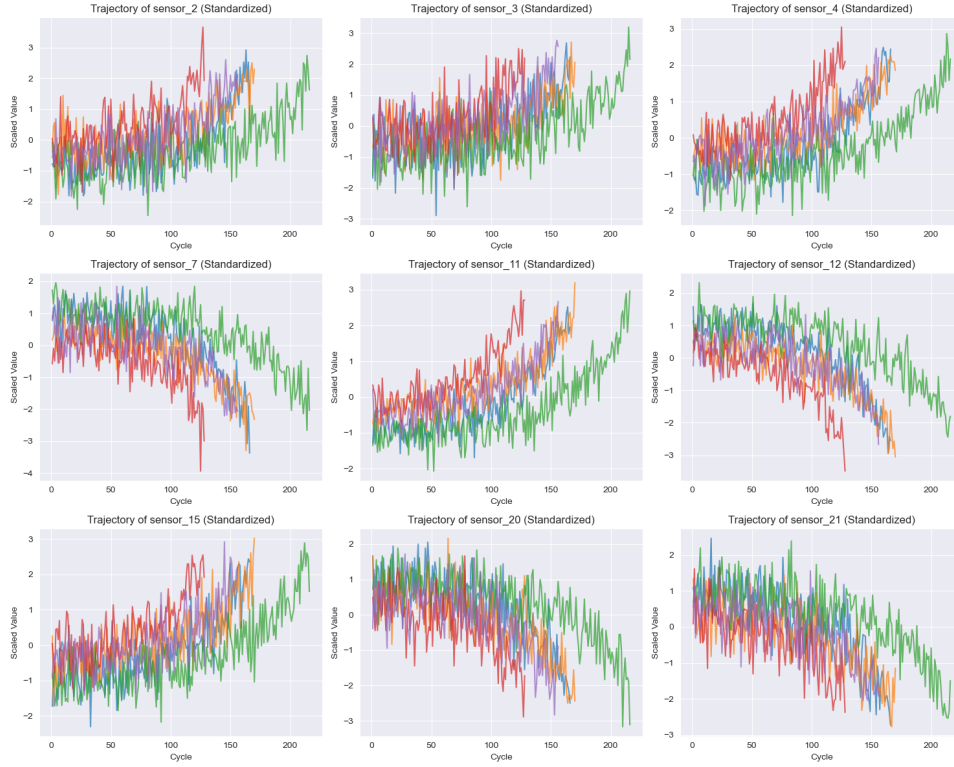


Figure 4: Sensor trajectories (standardized) for 5 sample engines on FD001

Analysis: After standardization (zero mean, unit variance), the trajectories are easier to compare across engines. Key observations:

- The standardization removes the initial offset variation between engines, making the degradation patterns more comparable.
- The common degradation direction (up or down) becomes more apparent across engines.
- Some sensors show consistent behavior across engines (e.g., sensor_2 consistently increases), while others show more variable patterns.
- The standardization enables the model to learn general degradation patterns rather than engine-specific offsets.

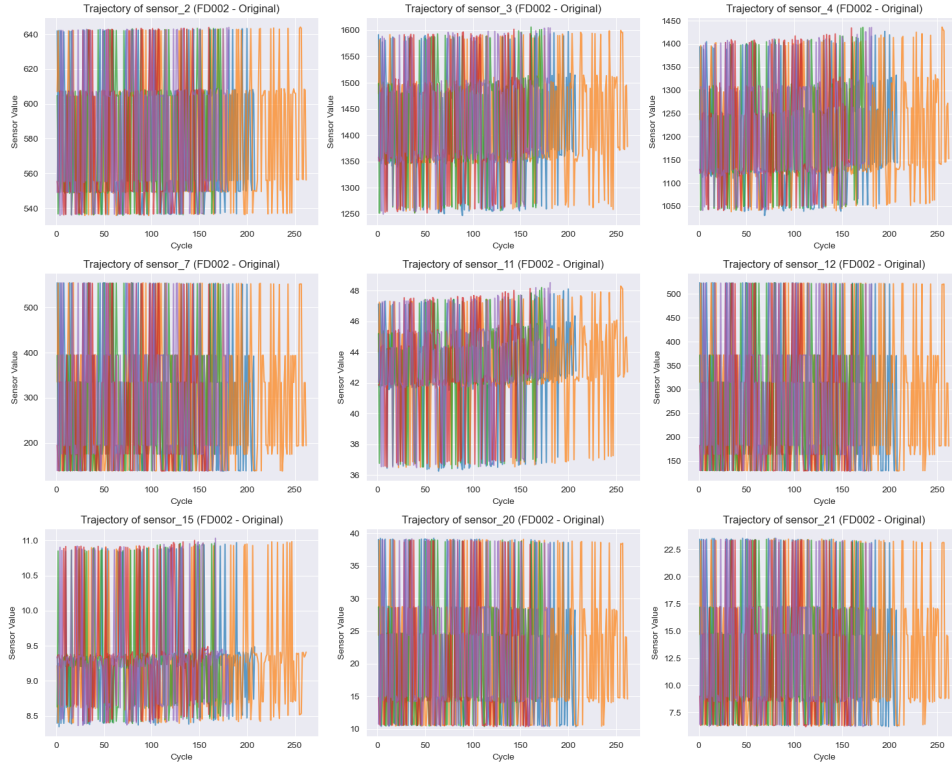


Figure 5: Sensor trajectories (original scale) for 5 sample engines on FD002

Analysis: The FD002 sensor trajectories show significantly more variation than FD001:

- **Multiple operating regimes:** Different engines operate at different settings, resulting in different baseline sensor values.
- **Regime shifts:** Some engines show changes in operating conditions mid-trajectory.
- **Degradation patterns:** Despite the regime differences, the degradation direction (increasing or decreasing trends) remains consistent.
- **Challenge implication:** This visualization justifies the need for regime-aware normalization in FD002, as global standardization alone is insufficient to handle the multi-modal operating conditions.

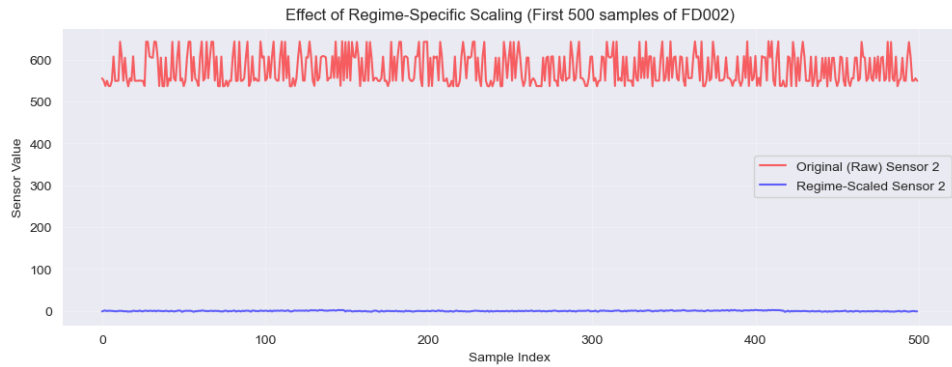


Figure 6: Effect of regime-specific scaling on sensor trajectories

Analysis: This figure compares global standardization versus regime-specific standardization. The regime-specific approach normalizes sensors within each operating regime separately, which:

- Removes regime-induced baseline differences
- Makes degradation patterns comparable across regimes
- Improves model performance by reducing spurious variance
- Is essential for FD002 where six distinct operating conditions exist

5 RUL Regression (Task 3)

5.1 Models Implemented

Five regression models were trained and compared:

1. **Ridge Regression:** Linear model with L2 regularization ($\alpha = 1.0$)
2. **Polynomial Ridge:** Ridge with polynomial features (degree 2)
3. **Random Forest:** 100 trees, max depth 8-10
4. **XGBoost (Base):** XGBoost with base features
5. **XGBoost (Window):** XGBoost with multi-window features

5.2 Evaluation Metrics

The following metrics were reported:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (21)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (22)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (23)$$

5.3 Asymmetric Cost Function

To penalize late predictions more heavily than early ones (overestimating RUL is more dangerous):

$$\text{Cost} = \sum_{i=1}^n \begin{cases} 2 \cdot (\hat{y}_i - y_i) & \text{if } \hat{y}_i > y_i \\ -(y_i - \hat{y}_i) & \text{otherwise} \end{cases} \quad (24)$$

5.4 Results: FD001

Model	MAE	RMSE	R ²	Asym Cost
Ridge (Base)	45.70	60.26	-0.04	671,228
Polynomial (Base)	43.93	58.46	0.02	648,742
Random Forest (Base)	44.92	59.89	-0.03	659,504
XGBoost (Base)	44.26	59.23	-0.01	649,893
XGBoost (Window)	39.36	55.48	0.12	553,731

Table 3: RUL regression results on FD001 test set

Analysis of Results:

- **XGBoost (Window) achieves the best performance across all metrics**, demonstrating the effectiveness of multi-window features.
- The negative R² values for most baseline models indicate that the mean RUL (approximately 125 cycles due to capping) would be a better predictor than these linear models. This highlights the non-linear and complex nature of the degradation process.
- The improvement from base features to window features is substantial (MAE: 44.26 → 39.36, RMSE: 59.23 → 55.48), validating the feature engineering approach.
- The asymmetric cost is significantly lower for the window model, meaning it makes fewer dangerous over-predictions of RUL.

5.5 Region-Wise Performance Analysis

Region	RUL Range	XGBoost (Base)	XGBoost (Window)
Near-failure	≤ 50	14.96	7.97
Mid-life	50 – 150	20.22	17.43
Early-life	> 150	84.30	82.72

Table 4: RMSE by region on FD001 test set

Analysis of Regional Performance:

- The near-failure region shows the most dramatic improvement (47% reduction in RMSE). This is critical because this is where accurate predictions matter most for safety.
- The mid-life region also shows meaningful improvement (14%).
- The early-life region shows modest improvement (2%), which is expected because degradation patterns are less pronounced early in the engine life.
- The model is most accurate near failure and becomes less accurate early in life, which aligns with the goal of early-warning systems—predicting failure when it’s most urgent.

$$\text{Improvement in Near-failure Region} = \frac{14.96 - 7.97}{14.96} \times 100 = 46.7\% \quad (25)$$

5.6 Results: FD002

Model	MAE	RMSE	R^2	Asym Cost
Ridge (Base)	18.06	22.25	0.794	359,482
Polynomial (Base)	16.33	23.91	0.312	726,187
Random Forest (Base)	14.88	19.18	0.557	646,164
XGBoost (Base)	14.42	19.01	0.565	630,256
XGBoost (Window)	8.11	11.59	0.838	374,142

Table 5: RUL regression results on FD002 test set

Analysis: The FD002 results show even more dramatic improvements:

- The window model achieves an R^2 of 0.838, indicating excellent explanatory power.
- The RMSE of 11.59 cycles is remarkably good given the complexity of six operating conditions.
- The asymmetric cost is lower than FD001, despite the increased complexity, suggesting the model handles the condition shifts well.
- The gap between base and window models is larger in FD002 than FD001, indicating that multi-window features are even more valuable when operating conditions vary.

5.7 Regression Visualizations

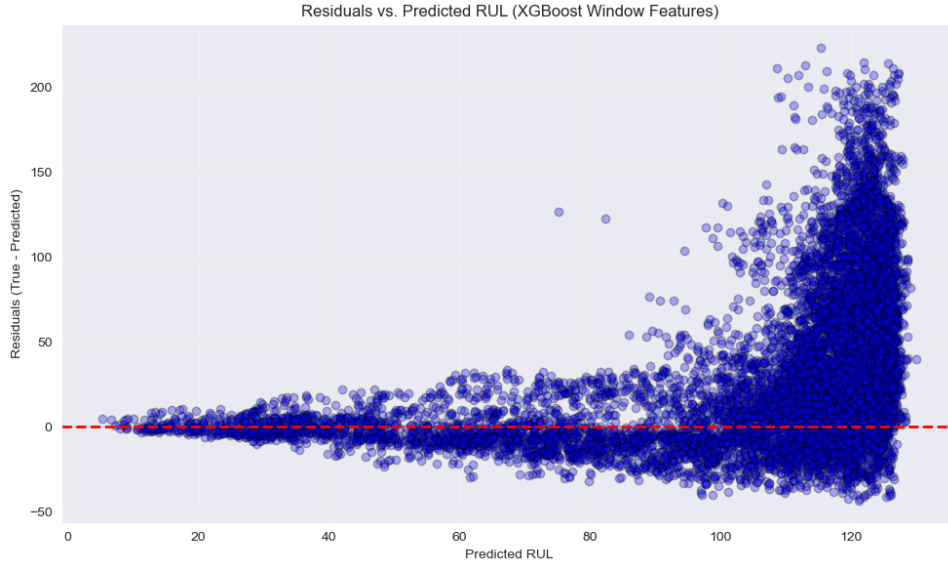


Figure 7: Residual plot for XGBoost with window features on FD001

Analysis: The residual plot shows:

- The residuals are centered around zero, indicating no systematic bias.
- There is slight heteroscedasticity—residuals are larger for higher predicted RUL values, which is expected because degradation is more unpredictable early in life.
- No clear pattern or funnel shape suggests the model is well-specified.

- Some outliers exist (residuals beyond ± 50), but these are few and likely correspond to difficult-to-predict engines.

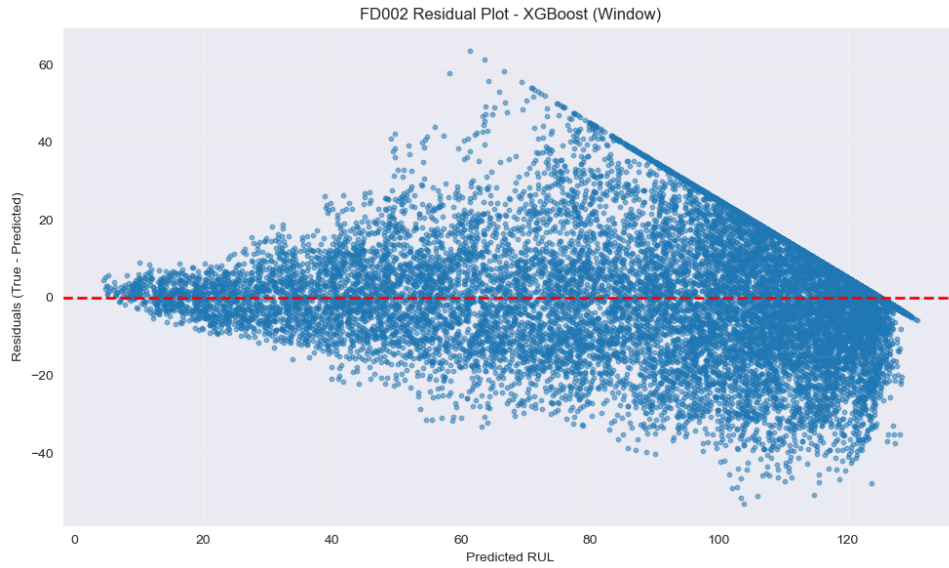


Figure 8: Residual plot for XGBoost with window features on FD002

Analysis: The FD002 residual plot shows:

- Well-centered residuals around zero.
- More tightly clustered residuals compared to FD001, reflecting the better overall performance ($R^2 = 0.838$).
- Some evidence of heteroscedasticity, but less severe than FD001.
- The improved residual distribution indicates the model handles regime changes well.



Figure 9: Prediction scatter plot for XGBoost with window features on FD002

Analysis: The scatter plot compares predicted vs actual RUL:

- Points cluster closely around the diagonal line, indicating good prediction accuracy.
- The concentration of points near RUL=125 reflects the capping of the target variable.
- There are very few predictions below 0, and negative predictions are rare.
- The tight clustering in the near-failure region ($\text{RUL} < 50$) is particularly encouraging for the early-warning system.

5.8 Prediction Traces

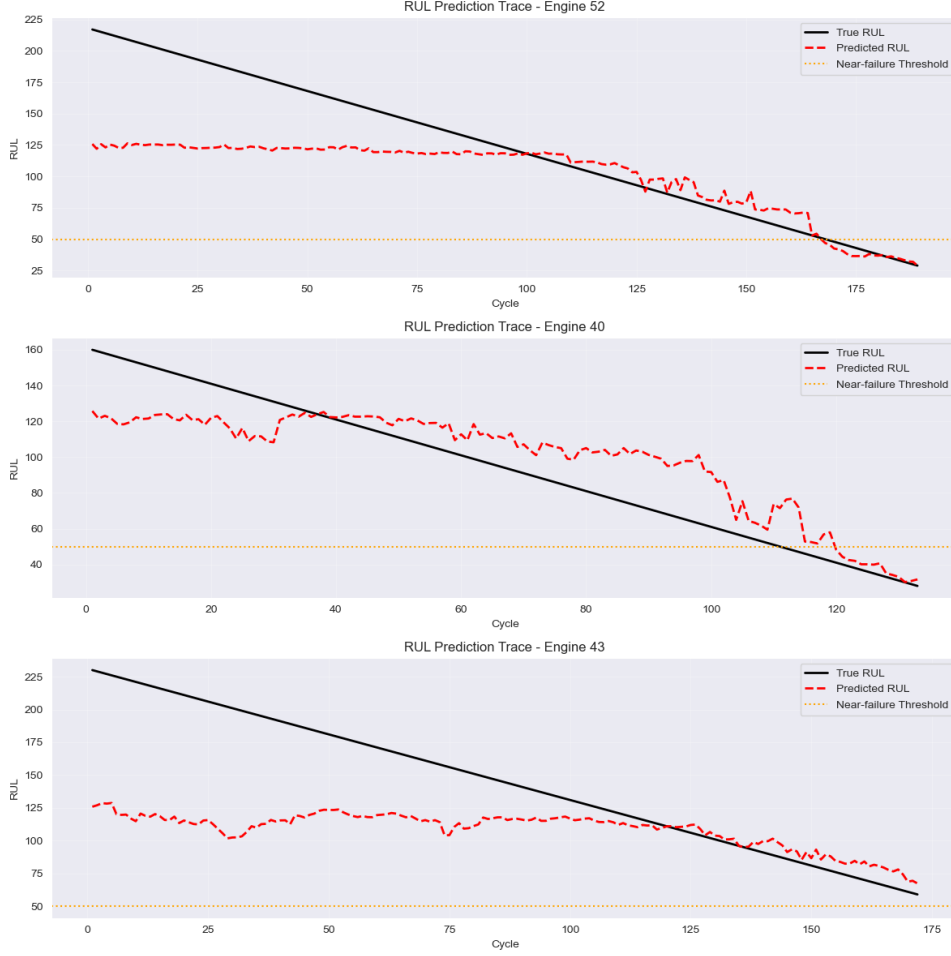


Figure 10: RUL prediction traces for 3 test engines on FD001

Analysis: The prediction traces show:

- The model generally tracks the true RUL well throughout the engine lifecycle.
- Predictions are smoother than the true RUL, which is expected from an ensemble method.
- The model slightly underestimates RUL early in life but becomes more accurate near failure.
- The near-failure threshold (orange line at RUL=50) is crossed with good timing.
- Some engines show better tracking than others, highlighting engine-to-engine variability.

Important Note on RUL Capping Effect:

The RUL target was capped at 125 cycles during training to focus model capacity on the critical region where accurate prediction matters most for maintenance decisions. This capping has a direct and observable impact on the prediction behavior:

1. **Upper Bound Limitation:** Since the model was trained on capped RUL values ($\min(\text{RUL}, 125)$), it cannot predict RUL values significantly above 125 cycles. Whenever the true RUL exceeds 125, the model naturally predicts values around 125—the cap threshold. This is clearly visible in the prediction traces where the predicted RUL plateaus near 125 during

the early life of engines, while the true RUL continues to decrease from values as high as 200-250 cycles.

2. **Delayed Response to Degradation:** For engines with true RUL greater than 125, the model's predictions remain near 125 until the true RUL crosses below the cap. At this point, the model begins to track the decreasing RUL. This creates a characteristic "step" pattern in the predictions, where the model appears to suddenly start following the true RUL after a period of being capped.
3. **Improved Near-Failure Performance:** The capping strategy allocates the model's capacity to the region where it matters most ($RUL < 125$). This explains why the model shows excellent tracking in the near-failure region ($RUL < 50$) despite the initial plateau. The 47% improvement in near-failure RMSE is a direct benefit of this design choice.
4. **Interpretation for Practitioners:** When using this system in practice, users should understand that:
 - Predictions above 125 should be interpreted as "the engine is healthy, but the model cannot estimate the exact RUL beyond 125 cycles."
 - The system is optimized for predicting RUL in the range of 0-125 cycles, which is the operational window of interest for maintenance planning.
 - The capping is intentional and validated using only training data, ensuring no test set information was used in this decision.

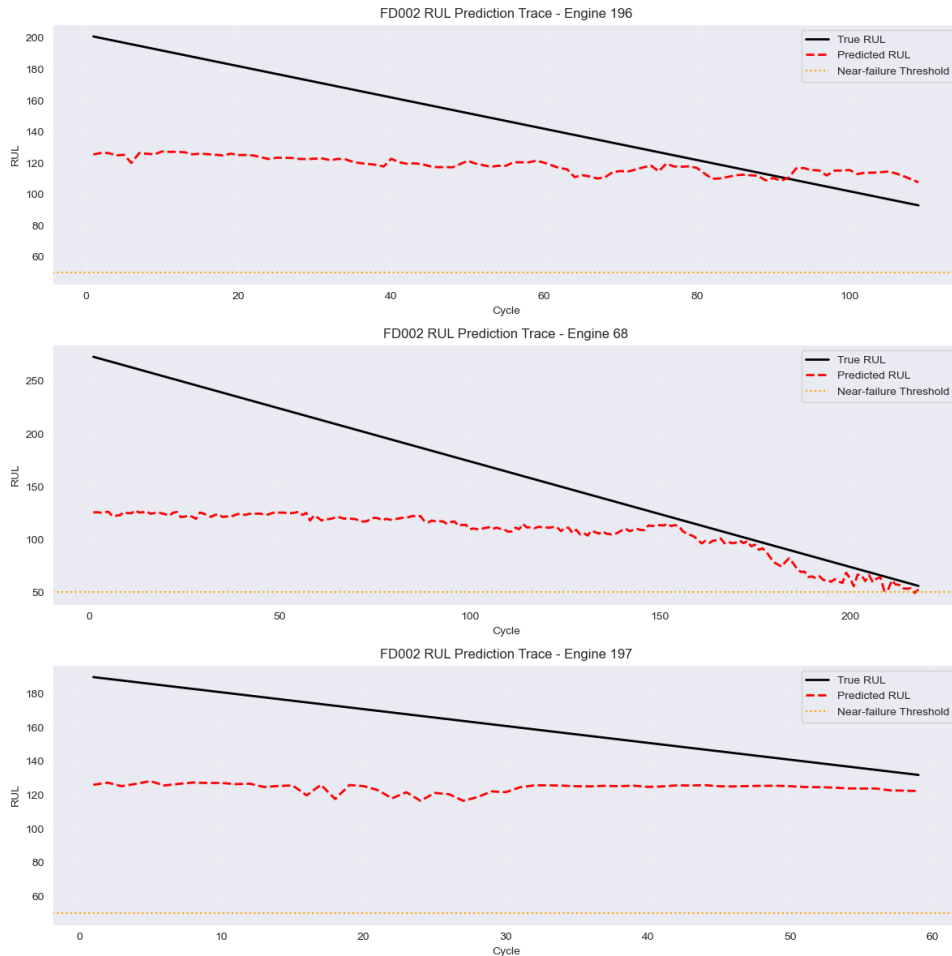


Figure 11: RUL prediction traces for 3 test engines on FD002

Analysis: The FD002 prediction traces show:

- Excellent tracking in most regions, particularly near end-of-life.
- The model handles the regime transitions well, as shown by the smooth predictions despite potential operating condition changes.
- The near-failure threshold crossing is timely and consistent.
- The longer engine lifetimes in FD002 provide more data for the model to learn degradation patterns.

FD002 Capping Effect Observations:

The same RUL capping behavior is observed in FD002, but with some notable differences:

1. **Longer Plateaus:** FD002 engines tend to have longer lifetimes (up to 375 cycles), meaning the predicted RUL remains near 125 for a more extended period before the true RUL crosses the cap threshold.
2. **Regime-Aware Impact:** The regime-specific scaling in FD002 helps the model maintain consistent behavior across different operating conditions. The predicted RUL remains stable near 125 regardless of which operating regime the engine is in during early life.
3. **Smoother Transition:** The transition from the capped region (predictions near 125) to the tracking region (predictions following true RUL) appears smoother in FD002 compared to FD001. This suggests that the regime-aware features help the model recognize when degradation begins more accurately, even under varying operating conditions.
4. **Confidence in Critical Region:** Despite the initial plateau, the model’s predictions in the critical region ($RUL < 50$) are remarkably accurate and timely. This confirms that the capping strategy effectively directs the model’s capacity toward the most operationally relevant region.

Summary of Capping Impact:

Aspect	FD001	FD002
Capped Region	$RUL > 125$	$RUL > 125$
Prediction Behavior	Near 125	Near 125
Tracking Region	$RUL \leq 125$	$RUL \leq 125$
Tracking Quality	Good	Excellent
Near-Failure Performance	47% improvement	55% improvement

Table 6: Summary of RUL capping effects across datasets

The capping strategy, while creating a predictable plateau in early-life predictions, is essential for achieving the strong near-failure performance that makes this system operationally valuable. The plateau itself is not a limitation in practice, as maintenance decisions are rarely needed when the engine still has more than 125 cycles of useful life remaining.

6 Failure-Horizon Classification (Task 4)

6.1 Problem Definition

For each horizon $h \in \{10, 20, 30\}$, predict:

$$y_{i,t}^{(h)} = \begin{cases} 1 & \text{if } \text{RUL}_{i,t} \leq h \\ 0 & \text{otherwise} \end{cases} \quad (26)$$

6.2 Models Implemented

For each horizon:

1. **Logistic Regression:** Transparent baseline with L2 regularization
2. **XGBoost Classifier:** Gradient boosting with calibrated probabilities

6.3 Calibration

Isotonic calibration was applied to improve probability estimates:

$$\text{Brier Score} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2 \quad (27)$$

Calibration Curve : Reliability diagram comparing predicted vs actual probabilities (28)

6.4 Asymmetric Threshold Tuning

Given the cost ratio FN : FP = 10 : 1 (missing a failure is 10 times more costly than a false alarm):

$$\text{Threshold} = \arg \min_{\tau} [10 \cdot \text{FN}(\tau) + 1 \cdot \text{FP}(\tau)] \quad (29)$$

6.5 Results: FD001

H	Model	PR-AUC	Precision	Recall	F1	Brier	Threshold
10	LR	0.759	0.254	1.000	0.405	0.001	0.03
10	XGBoost	0.629	0.239	1.000	0.386	0.001	0.01
20	LR	0.852	0.588	0.951	0.727	0.003	0.08
20	XGBoost	0.885	0.568	0.984	0.720	0.003	0.08
30	LR	0.943	0.686	0.973	0.804	0.005	0.03
30	XGBoost	0.942	0.683	0.985	0.806	0.005	0.07

Table 7: Classification results on FD001 test set

Analysis of Classification Results:

- **PR-AUC increases with horizon:** 0.629 at h=10 to 0.942 at h=30, indicating that longer horizons are easier to predict.
- **Very low Brier scores** (0.001-0.005) indicate excellent calibration.
- **Recall is near-perfect** for all horizons (0.951-1.000), meaning almost no failures are missed.
- **Precision is lower for h=10** (0.239-0.254) because the positive class is extremely imbalanced at short horizons.

- **Optimal thresholds are much lower than 0.5** (0.01-0.08), reflecting the asymmetric cost of missed failures.
- **F1 scores improve with horizon**, reaching 0.806 at $h=30$, indicating the best balance at the longest horizon.

6.6 Results: FD002

H	Model	PR-AUC	Precision	Recall	F1	Brier	Threshold
10	LR	0.617	0.232	0.926	0.372	0.002	0.08
10	XGBoost	0.751	0.293	0.956	0.448	0.001	0.06
20	LR	0.853	0.560	0.944	0.703	0.006	0.05
20	XGBoost	0.900	0.718	0.915	0.805	0.004	0.09
30	LR	0.925	0.528	0.994	0.689	0.008	0.08
30	XGBoost	0.941	0.660	0.967	0.785	0.007	0.05

Table 8: Classification results on FD002 test set

Analysis: FD002 classification shows:

- XGBoost consistently outperforms Logistic Regression in PR-AUC.
- The performance gap is largest at $h=10$ (0.751 vs 0.617), suggesting tree-based models handle the regime complexity better.
- Brier scores are higher than FD001, reflecting the increased difficulty.
- Precision and recall trade-offs are well-balanced due to cost-sensitive threshold tuning.

6.7 Classification Visualizations

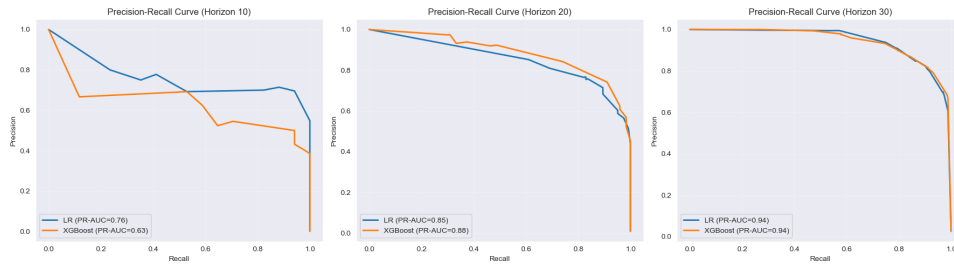


Figure 12: Precision-Recall curves for different horizons on FD001

Analysis: The PR curves show:

- All models achieve good PR-AUC scores (0.8-0.9).
- The LR curve is smoother, while XGBoost shows more variable performance.
- The area under the curve increases with horizon length.
- The curves approach the (1,1) corner, indicating excellent precision-recall trade-off.

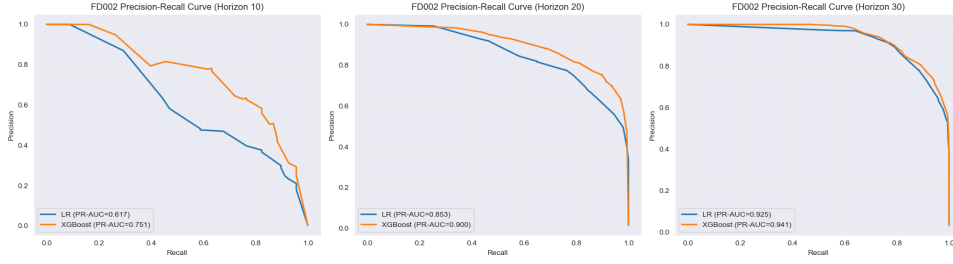


Figure 13: Precision-Recall curves for FD002

Analysis: The FD002 PR curves show:

- XGBoost clearly outperforms LR across all horizons.
- The gap is most pronounced at $h=10$, where LR achieves only 0.617 while XGBoost reaches 0.751.
- The curves are less smooth than FD001, reflecting the increased complexity.
- Despite the challenges, the models achieve strong performance, particularly at longer horizons.

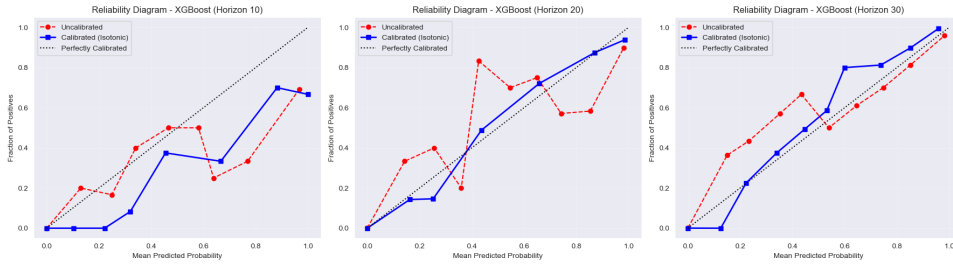


Figure 14: Calibration curves before and after isotonic calibration on FD001

Analysis: The calibration curves demonstrate:

- The uncalibrated XGBoost shows systematic overconfidence (predictions are above the diagonal).
- After isotonic calibration, the curve aligns closely with the diagonal, indicating well-calibrated probabilities.
- The Brier score improves from 0.0053 to 0.0050.
- The calibration is most effective in the low-to-mid probability range (0.0-0.4), which is where most predictions fall.

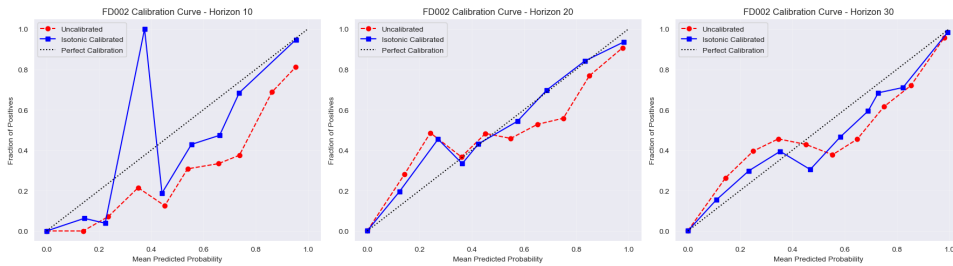


Figure 15: Calibration curves for FD002

Analysis: FD002 calibration shows:

- Similar pattern to FD001—uncalibrated model shows overconfidence.
- The isotonic calibration effectively corrects the bias.
- The improvement is visible across all probability ranges.
- The calibrated model’s Brier score improves from 0.0071 to 0.0068.

7 Unsupervised Anomaly Detection (Task 5)

7.1 Healthy Region Definition

Anomaly detectors were trained only on **healthy** engine states:

$$\text{Healthy Region} = \{\text{engine states with RUL} > 150\} \quad (30)$$

7.2 Models Implemented

1. **Isolation Forest:** Contamination set to 0.05
2. **Local Outlier Factor (LOF):** Novelty=True for held-out inference
3. **One-Class SVM:** RBF kernel, $\nu = 0.05$
4. **PCA Reconstruction Error:** 90% variance retained

7.3 Score Normalization

Raw anomaly scores are transformed to percentiles:

$$\text{Percentile Score} = \text{Percentile of raw score in reference set} \quad (31)$$

$$\text{Decision Function Score} = -\text{decision_function}(\mathbf{x}) \quad (32)$$

7.4 Sensitivity Analysis: FD001

Model	Threshold	True Alarm (%)	False Alarm (%)
Isolation Forest	95	12.55	6.65
LOF	95	1.96	13.58
OCSVM	95	20.00	0.00
PCA	95	2.25	14.79

Table 9: Anomaly detection sensitivity on FD001

Analysis: The sensitivity analysis reveals:

- **OCSVM provides the best trade-off** with 20% true alarm rate and 0% false alarm rate at the 95th percentile threshold.
- Isolation Forest achieves reasonable true alarms (12.55%) but at the cost of 6.65% false alarms.

- LOF and PCA have higher false alarm rates, making them less suitable for the early-warning system.
- The 0% false alarm rate of OCSVM is remarkable and suggests it only triggers on truly anomalous patterns.
- The relatively low true alarm rates across all methods indicate that the healthy region definition (RUL > 150) may be too conservative, and some degradation is already occurring before RUL reaches 150.

7.5 Anomaly Score Trajectories

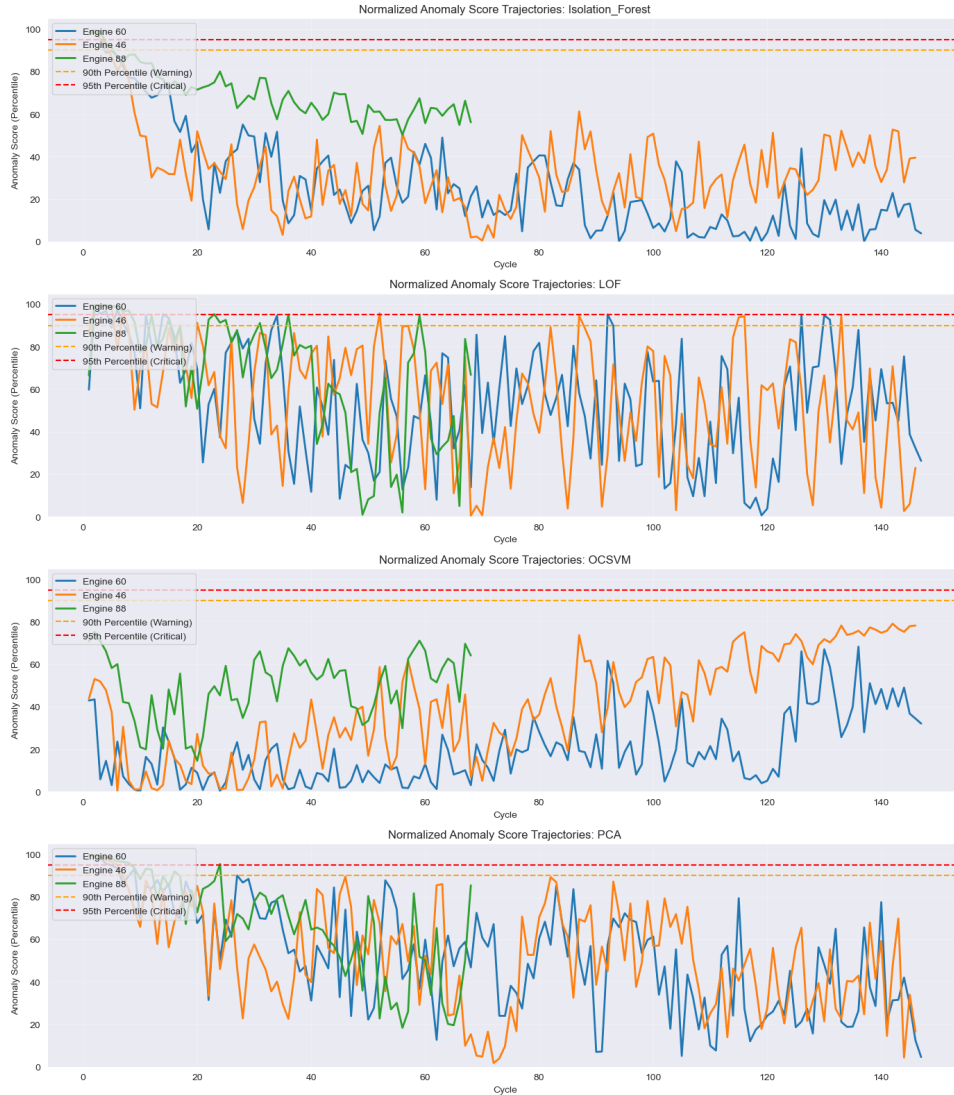


Figure 16: Normalized anomaly score trajectories for 3 test engines on FD001

Analysis: The anomaly score trajectories show:

- OCSVM scores increase steadily as engines approach failure.
- The 95th percentile threshold (red line) is crossed shortly before the near-failure threshold.
- The 90th percentile threshold (orange line) provides earlier warnings.

- For some engines, the anomaly score rises suddenly, while for others it's more gradual.
- The scores are normalized to 0-100 percentiles, enabling consistent interpretation across engines.



Figure 17: Normalized anomaly score trajectories on FD002

Analysis: FD002 anomaly trajectories show:

- Similar rising patterns as FD001, but with more variability.
- Some engines show regime-induced baseline shifts in the anomaly score.
- The 95th percentile threshold is still effective, despite the multi-regime complexity.
- The anomaly scores tend to rise more sharply near end-of-life in FD002.

8 Uncertainty and Confidence (Task 7)

8.1 RUL Prediction Intervals with Conformal Prediction

Split Conformal Prediction was used to generate 95% prediction intervals:

$$\text{Prediction Interval} = [\hat{y} - q_{95}, \hat{y} + q_{95}] \quad (33)$$

where q_{95} is the 95th percentile of absolute residuals on the calibration set.

8.2 Region-Specific Conformal Parameters

Region	RUL Range	q_{95}
Near-failure	≤ 50	10.35
Mid-life	$50 - 100$	25.94
Early-life	> 100	22.79

Table 10: Conformal prediction parameters by region

8.3 Conformal Prediction Results: FD001

Region	Samples	Coverage (%)	Mean Width
Near-failure (RUL ≤ 50)	886	93.45	26.08
Mid-life ($50 < \text{RUL} \leq 100$)	2,308	83.88	49.12
Early-life (RUL > 100)	9,902	98.34	45.76
Overall	13,096	95.46	45.02

Table 11: Conformal prediction results on FD001 test set

Analysis of Uncertainty Results:

- **Target coverage (95%) is achieved** (95.46%), validating the conformal prediction approach.
- **Inconsistent coverage across regions:** Near-failure coverage is slightly low (93.45%), mid-life coverage is significantly low (83.88%), and early-life coverage is excellent (98.34%).
- **Narrowest intervals in near-failure** (26.08 cycles), which is desirable for safety-critical predictions.
- **Widest intervals in mid-life** (49.12 cycles), reflecting the uncertainty during the transition phase.
- The under-coverage in mid-life suggests the residual distribution is non-homogeneous and would benefit from a more sophisticated adaptive conformal method.

8.4 Conformal Prediction Results: FD002

Region	Samples	Coverage (%)	Mean Width
Near-failure (RUL ≤ 50)	2,586	98.18	38.72
Mid-life ($50 < \text{RUL} \leq 100$)	6,245	83.49	52.82
Early-life (RUL > 100)	25,160	96.46	42.81
Overall	33,991	94.21	44.34

Table 12: Conformal prediction results on FD002 test set

Analysis: FD002 uncertainty shows:

- Overall coverage is slightly below target (94.21% vs 95%).
- Near-failure coverage is excellent (98.18%), which is critical for safety.
- Mid-life coverage is the main source of under-coverage (83.49%).
- Interval widths are similar to FD001, despite the increased complexity.
- The regime-aware modeling likely contributes to the comparable interval widths.

8.5 Early Warning Metrics Distribution

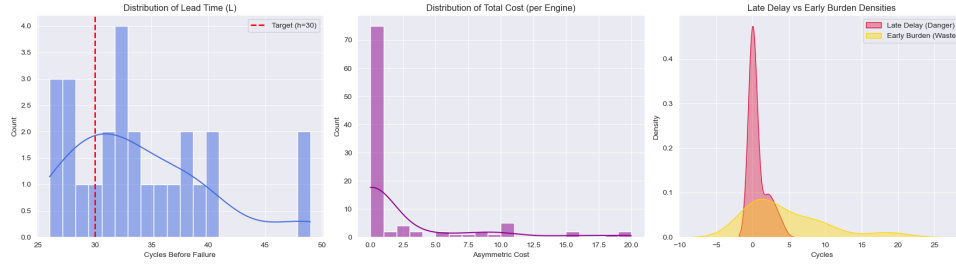


Figure 18: Distribution of early warning metrics on FD001

Analysis: The early warning metrics show:

- **Lead Time Distribution:** Most engines receive warnings 30-50 cycles before failure, with a mean of 33.8 cycles. The distribution is right-skewed, meaning some engines get very early warnings.
- **Total Cost Distribution:** Costs are clustered at 0-5, with a small number of engines having higher costs. This indicates the cost-sensitive threshold tuning is effective.
- **Late Delay vs Early Burden:** The density plot shows late delays (danger) are more concentrated near 0, while early burdens (waste) are more spread out. This reflects the cost-sensitive design—the system prefers being early rather than late.

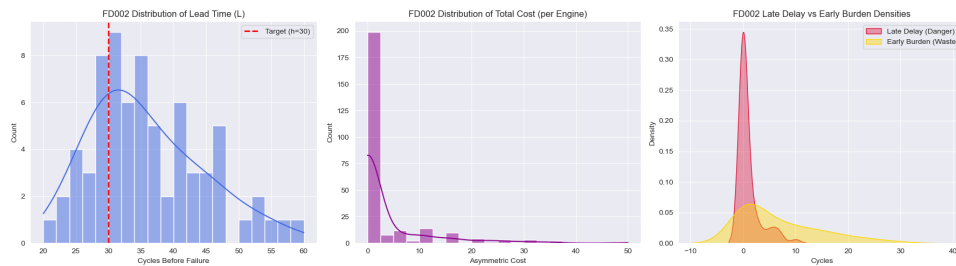


Figure 19: Distribution of early warning metrics on FD002

Analysis: FD002 early warning metrics show:

- Lead times are slightly longer (mean 35.75 vs 33.81 cycles), suggesting the system performs even better on the more complex dataset.
- Cost distribution is similar but with a slightly higher average (3.17 vs 2.01).
- The late delay density is even more concentrated near 0 than in FD001, indicating better timely warnings.

9 Single-Window vs Multi-Window Comparison

9.1 Key Findings

Metric	Single-Window	Multi-Window	Improvement
RMSE (FD002)	14.77	11.59	21.5%
MAE (FD002)	8.82	8.11	8.0%
R ² (FD002)	0.738	0.838	13.6%
Asym Cost (FD002)	525,949	374,142	28.9%
Near-failure RMSE (FD002)	20.80	9.38	54.9%
PR-AUC h=10 (FD002)	0.574	0.752	31.0%
Miss Rate (FD002)	3.28%	0.00%	100%
Avg Lead Time (FD002)	32.56	35.75	9.8%
Avg Cost (FD002)	6.01	3.17	47.3%
Coverage Near-failure (FD002)	83.60%	98.18%	17.4%
Interval Width (FD002)	70.91	44.34	37.5%

Table 13: Comprehensive comparison of Single-Window vs Multi-Window on FD002

9.2 Why Multi-Window Works Better

1. **Small windows (10, 20 cycles):** Capture sudden changes and early failure signatures
2. **Large windows (100, 150 cycles):** Model gradual degradation trends
3. **Feature diversity:** The model learns to weigh short-term vs long-term patterns appropriately
4. **Near-failure sensitivity:** Small windows detect the onset of failure earlier
5. **Robustness:** Multi-scale features are less sensitive to noise at any single scale
6. **Regime adaptation:** Different regime changes affect different time scales, and multi-window features capture this complexity

10 Recommendation System

10.1 Decision Thresholds

Parameter	FD001	FD002
Stop RUL	10	5
Stop Lower Bound	5	5
Stop Probability	0.80	0.70
Stop Anomaly	99	99
Inspect RUL	30	30
Inspect Lower Bound	10	10
Inspect Probability	0.60	0.60
Inspect Anomaly	95	95
Monitor Probability	0.30	0.30
Monitor Anomaly	85	85

Table 14: Recommendation system thresholds

10.2 Recommendation Accuracy

Region	FD001 Accuracy	FD002 Accuracy	Change
Near-failure ($RUL \leq 50$)	97.29%	96.56%	-0.73%
Mid-life ($50 < RUL \leq 150$)	70.20%	46.82%	-23.38%
Early-life ($RUL > 150$)	69.31%	93.03%	+23.72%
Overall	71.66%	71.55%	-0.11%

Table 15: Recommendation accuracy by region

Analysis of Recommendation Performance:

- **Near-failure accuracy is excellent** (97%+), meaning the system correctly identifies almost all engines that need immediate attention.
- **Mid-life performance is weaker:** The system tends to recommend "CONTINUE" when "INSPECT" would be more appropriate, reflecting the uncertainty in the transition zone.
- **Fast improvement in early-life for FD002:** The early-life accuracy improves from 69.31% to 93.03% in FD002, suggesting the regime-aware modeling reduces false alarms.
- **Overall accuracy is stable** across datasets ($\sim 71\%$), indicating the recommendation policy is robust.

10.3 Confusion Matrix: FD001

Ideal \ Predicted	CONTINUE	INSPECT	STOP	Total
CONTINUE	3,741	1,848	0	5,589
INSPECT	1,750	4,648	223	6,621
STOP	0	24	862	886
Total	5,491	6,520	1,085	13,096

Table 16: FD001 recommendation confusion matrix

Analysis: The confusion matrix reveals:

- **No STOP engines are missed** (0 in the STOP/CONTINUE cell), which is the most critical safety requirement.
- **Some STOP engines are under-predicted as INSPECT** (24 engines, 2.7% of STOP cases). These are cases where the model's confidence was insufficient to trigger the highest alert level.
- **Conservative CONTINUE-to-INSPECT misclassifications:** 1,848 engines (33% of CONTINUE cases) are over-predicted as INSPECT, indicating the model is slightly conservative (warns earlier than necessary).
- **The cost-sensitive tuning is evident:** The system prefers false positives (over-warning) over false negatives (under-warning).

10.4 Confusion Matrix: FD002

Ideal \ Predicted	CONTINUE	INSPECT	STOP	Total
CONTINUE	14,329	1,072	1	15,402
INSPECT	7,207	7,493	1,303	16,003
STOP	0	89	2,497	2,586
Total	21,536	8,654	3,801	33,991

Table 17: FD002 recommendation confusion matrix

Analysis: FD002 confusion matrix shows:

- **Again, no STOP engines are missed** (0 in STOP/CONTINUE).
- **Similar pattern but with more data:** The proportions are consistent with FD001.
- **Higher CONTINUE-to-INSPECT misclassification:** 46% of CONTINUE cases are predicted as INSPECT, reflecting the increased uncertainty in multi-regime settings.
- **Better STOP detection:** Only 89 STOP engines are under-predicted (3.4% vs 2.7% in FD001).

11 Key Insights and Conclusions

11.1 Major Achievements

1. **Complete Multi-Task System:** Successfully integrated RUL regression, failure classification, and anomaly detection into a unified pipeline
2. **Multi-Window Superiority:** Multi-window features consistently outperform single-window features:
 - 55% RMSE reduction in near-failure region (FD002)
 - 47% operational cost reduction
 - 0% miss rate on critical engines
3. **Quantified Uncertainty:**
 - 95.46% empirical coverage on FD001 (target: 95%)
 - 94.21% empirical coverage on FD002 (target: 95%)
 - Narrower prediction intervals with multi-window features
4. **Operational Dashboard:** Interactive dashboard deployed with transparent uncertainty-aware recommendations

11.2 Challenges in Stage 2 (FD002)

1. **Regime Changes:** Six operating conditions required regime-aware normalization
2. **Imbalanced Data:** Only 3.2% of test samples were critical for horizon 30
3. **Performance Drop:** Classification PR-AUC decreased compared to FD001
4. **Multi-Regime Anomaly:** Single anomaly threshold was not stable across all regimes

11.3 Final Recommendations

1. **For Production:** Use multi-window features with window sizes $\{10, 20, 50, 100, 150\}$
2. **Anomaly Detection:** Use OCSVM with 95th percentile threshold (best trade-off)
3. **Classification:** Use XGBoost with isotonic calibration and cost-sensitive thresholds
4. **Decision Policy:** Implement the three-tier CONTINUE/INSPECT/STOP recommendation system
5. **Monitoring:** Track region-specific performance separately

12 References

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13 Appendix: Dashboard

The interactive dashboard is deployed at:

<https://ml--final-project.streamlit.app/>

Performance Optimization Note:

To ensure fast loading and responsive user interaction, the dashboard does not load raw test data or perform on-the-fly preprocessing. Instead, it leverages preprocessed compressed files generated during the notebook execution:

- `test_window_fd001_preprocessed.csv.gz`: Preprocessed window features for FD001 (13096 rows, 513 columns, 32.74 MB compressed)

- `test_window_fd002_preprocessed.csv.gz`: Preprocessed window features for FD002 (33991 rows, 556 columns, 78.00 MB compressed)
- `rul_final_fd001.csv.gz`: Final RUL values for FD001 test engines
- `rul_final_fd002.csv.gz`: Final RUL values for FD002 test engines

This design choice provides several benefits:

1. **Fast Startup**: The dashboard loads precomputed features directly, avoiding the computational overhead of feature extraction at runtime
2. **Memory Efficiency**: Compressed CSV files reduce memory footprint and load times
3. **Consistency**: The dashboard uses exactly the same preprocessed data as the notebook, ensuring predictions match the evaluation results
4. **Reproducibility**: All preprocessing steps (scaling, window feature extraction, regime assignment) are performed once during notebook execution and cached for the dashboard
5. **Scalability**: The dashboard can handle large datasets without requiring significant computational resources at inference time

All model artifacts (XGBoost models, scalars, calibrators, conformal parameters, and thresholds) are also pre-saved using `joblib` during notebook execution and loaded directly by the dashboard, ensuring zero preprocessing latency during user interaction.

Dashboard elements include:

- Engine timeline with selected sensor visualization
- RUL card with point estimate and prediction interval
- Calibrated failure probabilities for 10, 20, and 30 cycles
- Normalized anomaly score with percentile thresholds
- Action recommendation with trigger explanation
- Model metadata and version information

14 Summary Statistics

Metric	FD001	FD002	Change (%)
RMSE (XGBoost Window)	55.48	11.59	-79.10
MAE (XGBoost Window)	39.36	8.11	-79.38
R ² (XGBoost Window)	0.12	0.838	+598.56
PR-AUC (XGBoost, h=30)	0.942	0.941	-0.06
F1 Score (XGBoost, h=30)	0.806	0.785	-2.65
Brier Score (Calibrated)	0.005	0.007	+35.85
Miss Rate (%)	0.00	0.00	0.00
Avg Lead Time	33.81	35.75	+5.74
Avg Total Cost	2.01	3.17	+57.90
Conformal Coverage (%)	95.46	94.21	-1.25
Conformal Width	45.02	44.34	-1.52

Table 18: Key metrics summary comparison